

Quantum Circuit Optimization with LLMs under a Structured Guideline

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WISA, August 27, 2026

- ➊ Introduction
- ➋ Contribution
- ➌ LLM-Based Quantum Circuit Optimization
- ➍ Results
- ➎ Concluding Remarks (and Note on Limitation)

Quantum Computing Land Landscape: 2021 and 2026

Advances in quantum computing

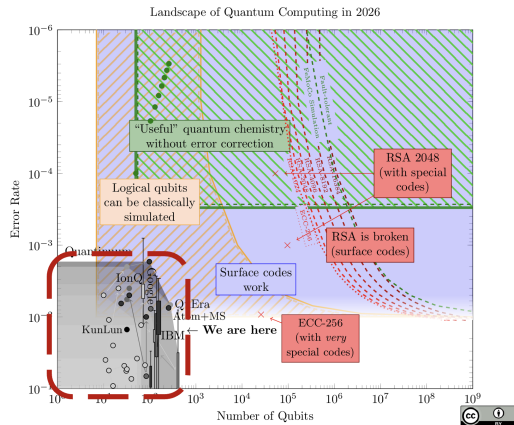
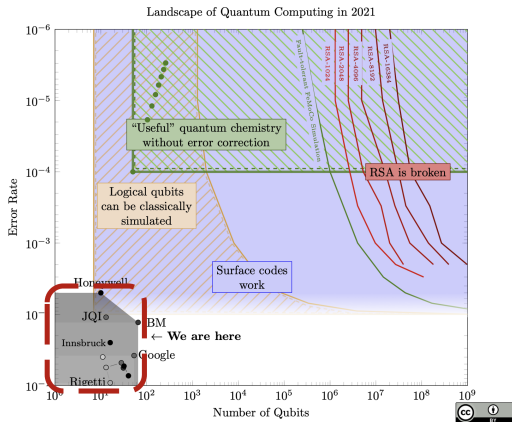


Figure 1: Quantum computing landscape¹ in 2021 (left) and 2026 (right).

¹Samuel Jaques, Quantum Landscape, https://sam-jaques.appspot.com/quantum_landscape

Quantum Computing Land Landscape: 2021 and 2026

“Attacks always get better” [Sch09]¹ (see “←--”)

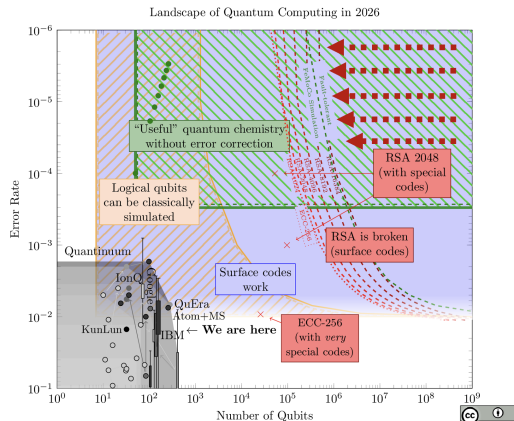
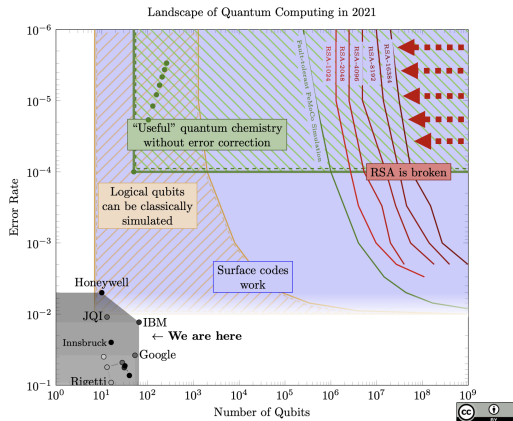


Figure 1: Quantum computing landscape in 2021 (left) and 2026 (right).

¹Bruce Schneier. New Attack on AES, 2009.

Quantum Computing Land Landscape: 2021 and 2026

Attack difficulty (easiest first): ECC → RSA →→ ... → AES, SHA-2/3

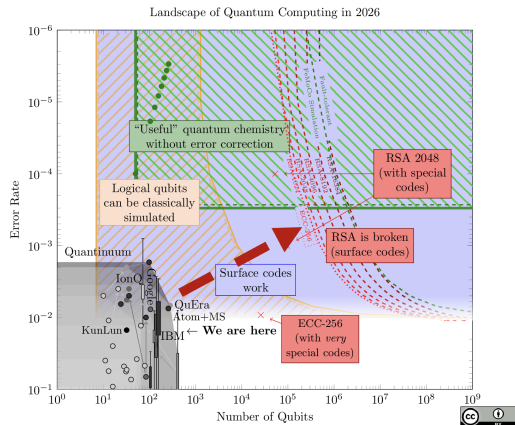
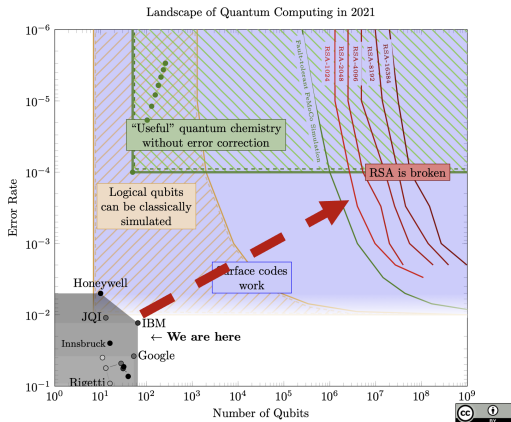


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 - Shor's algorithm for **RSA and ECC**
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e.g., AES, SHA-2/3, CHAM, HIGHT, LEA, LSH

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 - ECC $\Rightarrow$ Point addition
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Grover's Algorithm on Symmetric Key Cryptography

- The cost of Grover's algorithm is dominated by the **quantum implementation of the target cipher**.
⇒ Many works optimize quantum circuits for various target ciphers.

Quantum Analysis of AES
Kyungbae Jang¹, Anubhab Baksi^{a,2}, Hyunji Kim¹,
Gyeongju Song¹, Hwajeong Seo^{b,1} and Anupam Chattopadhyay³
¹ Hansung University, Division of IT Convergence Engineering
² Lund University, Elektro- och informationsteknik,
³ Nanyang Technological University, School of Computer Science

**New Results in Quantum Analysis of LED:
Featuring One and Two Oracle Attacks
New Results with Optimizations and Consideration for Modes**
Siyi Wang¹, Kyungbae Jang², Anubhab Baksi¹,
a Chakraborty³, Bryan Lee¹, Anupam Chattopadhyay¹ and
Hwajeong Seo²
¹ Nanyang Technological University, Singapore
² Hansung University, Seoul, Republic of Korea
³ Techno International New Town, Kolkata, India

**Grover on KATAN: Quantum Resource
Estimation**
MOSTAFIZAR RAHMAN[✉] AND GOUTAM PAUL[✉] (Senior Member, IEEE)
Cryptography and Security Research Unit, R. C. Bose Centre for Cryptology and Security, Indian Statistical Institute, Kolkata 700098,
India
Corresponding author: Goutam Paul (e-mail: goutam.k.paul@gmail.com).

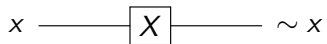
**Quantum Circuit Implementation and Resource
Analysis of LBlock and LiCi**
XiaoYu Jing¹, YanJu Li^{1,2}, GuangYue Zhao¹, Huiqin Xie^{1*}
¹ Beijing Electronic Science and Technology Institute, Beijing 100070, China
² Guangxi Key Laboratory of Cryptography and Information Security, Guilin University
of Electronic Technology, Guilin 541004, China

AES is implemented (in quantum) inside the Grover *oracle* and repeated.

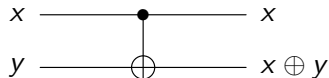


- For k -bit key, Grover repeats the oracle about $\sqrt{2^k}$ times
- Make that circuit **smaller and faster** $\Rightarrow$ the whole attack gets cheaper

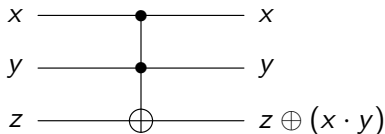
- There are several quantum gates to implement **Boolean operations**, such as:



(a) **X gate** $\Rightarrow$ NOT operation



(b) **CNOT gate** $\Rightarrow$ XOR operation



(c) **Toffoli gate** $\Rightarrow$ AND operation

- ① Introduction
- ② Contribution
- ③ LLM-Based Quantum Circuit Optimization
- ④ Results
- ⑤ Concluding Remarks (and Note on Limitation)

- We present an LLM-based framework for quantum circuit optimization
 - Replace manual expert implementation
- Guideline (.md) for verified and optimized quantum circuits
 - Collect general techniques that can be reused across different ciphers (e.g., work-qubit reuse)
- Evaluate our framework (i.e., LLM-based quantum circuit optimization)
 - New quantum implementations
 - ⇒ CRAFT, MANTIS, QARMA, mCrypton, EPCBC, and Pyjamask.
 - Compare with existing implementations
 - ⇒ LED [1], PRESENT, GIFT [2], and PRINCE [3].

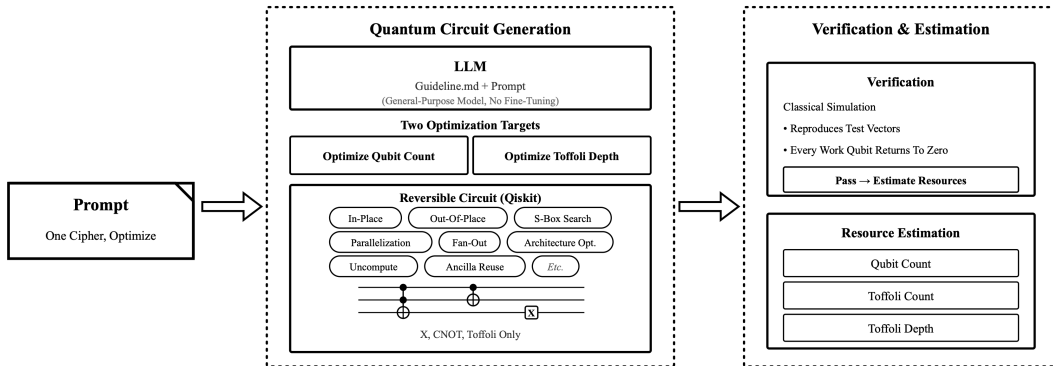
[1] S. Wang et al. "New results in quantum analysis of led: Featuring one and two oracle attacks", Indocrypt, 2025.

[2] K. Jang et al. "Efficient implementation of PRESENT and GIFT on quantum computers", Applied Sciences, 2021.

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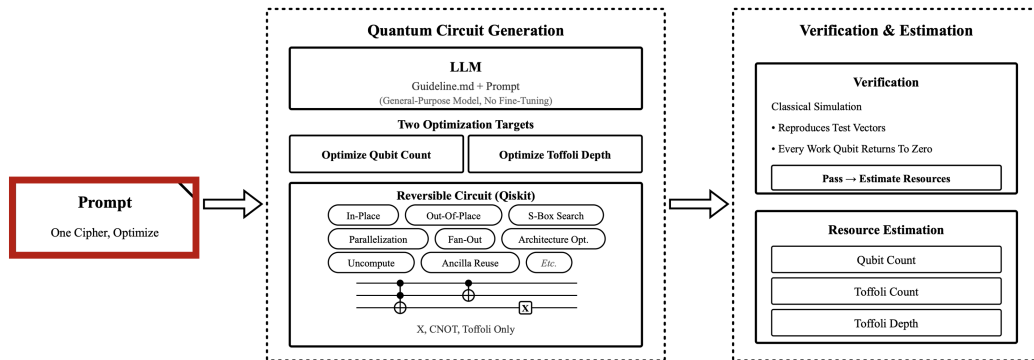
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Proposed Framework: Overview



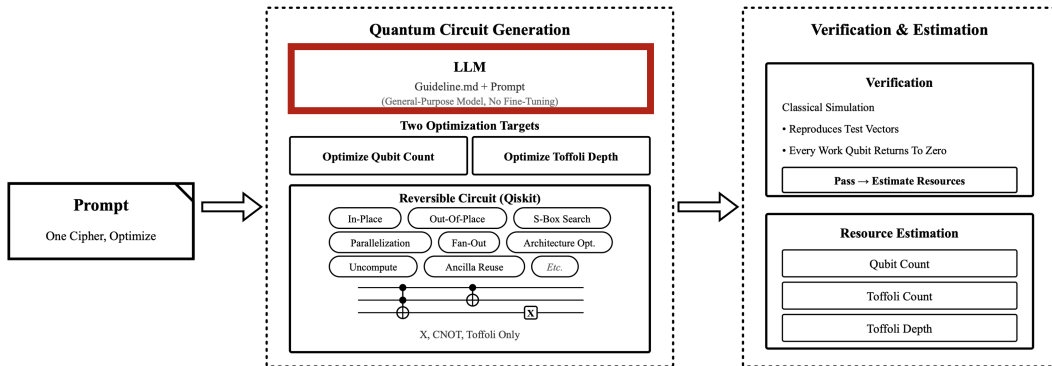
Target Cipher Selection

- We choose **only one cipher per task** for quantum circuit implementation.
⇒ **Multi-task prompting degrades LLM performance** [4], and we observed the same effect in quantum circuit implementation



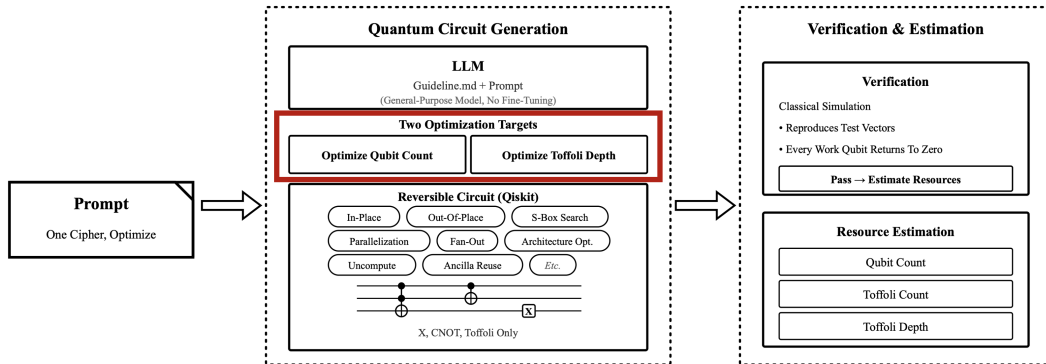
[4] J. Chen et al., "Understanding llm performance degradation in multi-instance processing: The roles of instance count and context length", arXiv preprint, arXiv:2603.22608, 2026.

- We include the guideline `Guideline.md` for LLM-based quantum implementation.
 - **Essential things in quantum circuit implementation** for cryptanalysis (more precisely, for Grover's attack).



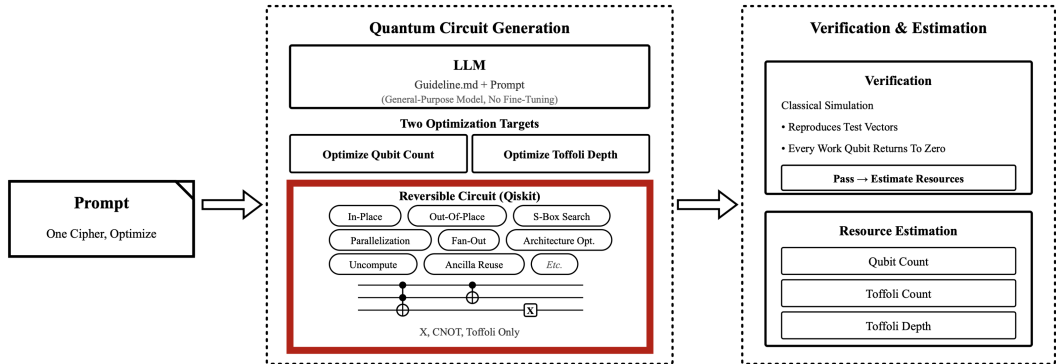
Optimization Strategy

- We specify two optimization strategies considering the main metrics in quantum implementations.
 - **Qubit optimization** $\Rightarrow$ Space complexity
 - **Depth optimization** $\Rightarrow$ Time complexity



Quantum Implementation Techniques

- Depending on the optimization strategy, the LLM selects the techniques well suited to that goal.
 - We collected generally applicable techniques for quantum implementations of cryptographic algorithms.



- Two main approaches for quantum implementations:
 - ① **In-place** implementation without separate output qubits; $|x\rangle \longrightarrow |f(x)\rangle$
 $\Rightarrow$ **Save qubits**, but **high circuit depth**
 - ② **Out-of-place** implementation with a separate output; $|x\rangle |0\rangle \longrightarrow |x\rangle |f(x)\rangle$
 $\Rightarrow$ Require **extra qubits** for output, but **low circuit depth**

Example: In-Place vs Out-of-Place

- The same outputs computed in two ways
 - Trade-off: **qubit count** $\leftrightarrow$ **Toffoli depth**

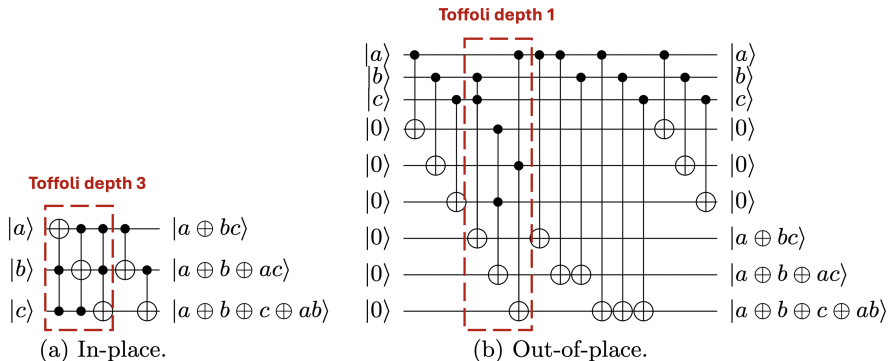


Figure 2: In-place (Toffoli depth 3) vs out-of-place (Toffoli depth 1).

- Ancilla qubits can be reused through **uncomputation** ($f^\dagger$).
 - Reverse the previous operation** to reset the ancilla qubits to a clean state (i.e., $|0\rangle$). $\Rightarrow$ Reduce **qubit count**, but **increase depth**

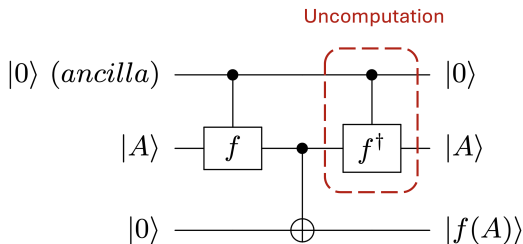


Figure 3: Cleaning a qubit through uncomputation ($f^\dagger$)

Techniques in Guideline: Balancing Qubit and Depth

- Optimizing **too heavily for one side** (qubits or depth) often leads to poor performance.
 - Figure (a): reuse single ancilla set $\Rightarrow$ depth $\times 3$
 - Figure (b): add 2 ancilla sets $\Rightarrow$ depth $\times 1$

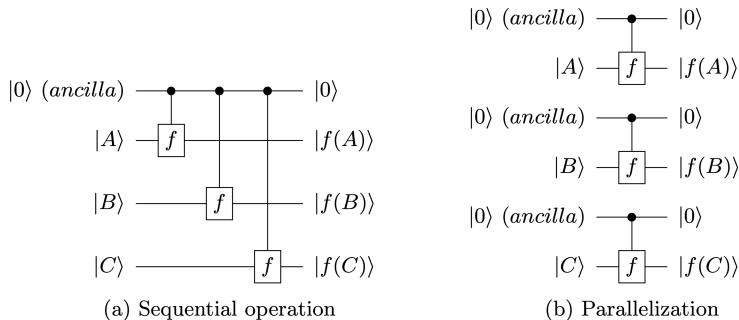


Figure 4: Reuse of a single ancilla versus multiple ancillas

Techniques in Guideline: Architecture Optimization

- We add the *Shallow* technique from [5] to the guideline.
⇒ Effective for the round-based encryption of block ciphers, and **generic!!**

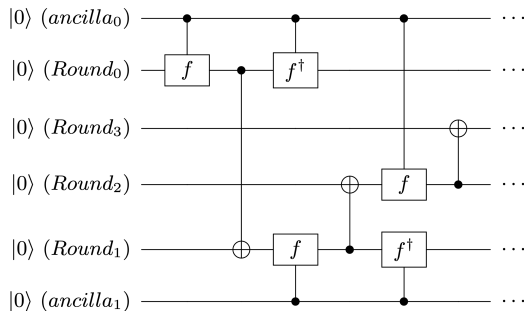


Figure 5: Shallow architecture in [5] (used for AES implementation)

[5] K. Jang et al., "Quantum Analysis of AES", IACR Communications in Cryptology, 2025.

- Reuse two ancilla sets ($ancilla_0$, $ancilla_1$) to reduce the depth overhead from uncomputation.

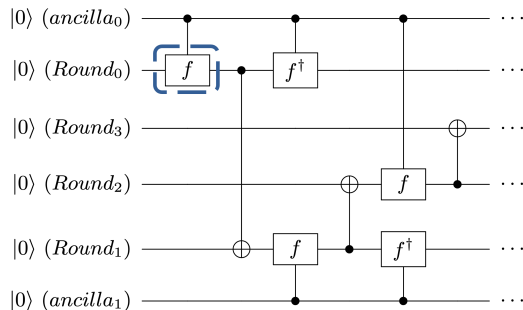


Figure 5: Shallow architecture (Round₀)

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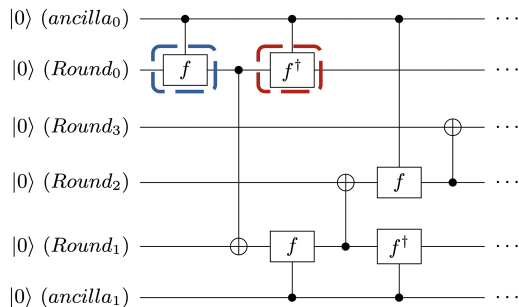


Figure 5: Shallow architecture ($Round_0 + Round_0^\dagger$)

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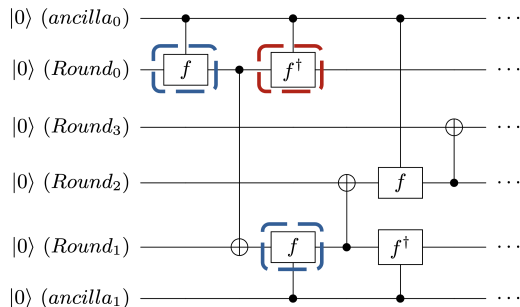


Figure 5: Shallow architecture ($Round_0 + Round_0^\dagger + Round_1$)

- Reuse two ancilla sets ($ancilla_0, ancilla_1$) to reduce the depth overhead from uncomputation.

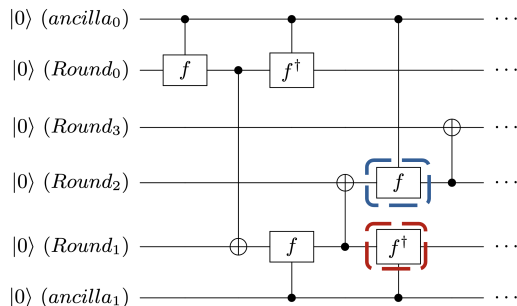
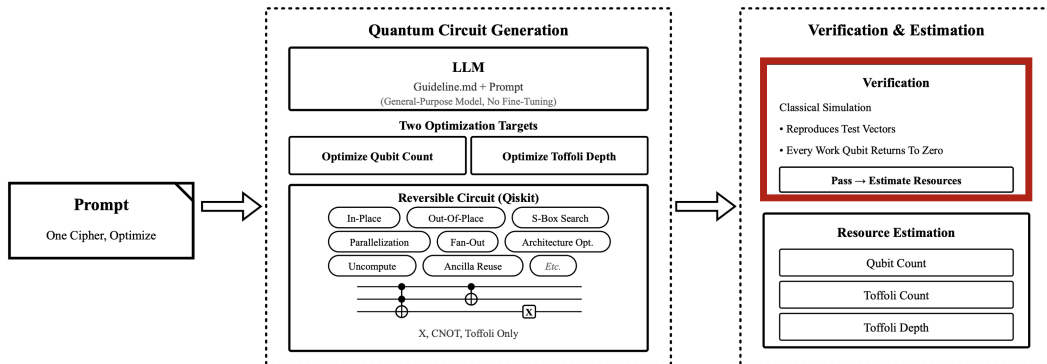
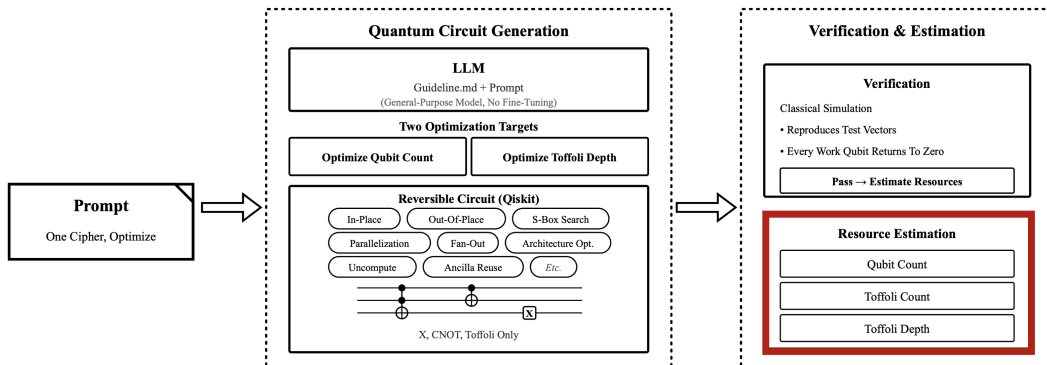


Figure 5: Shallow architecture ($\text{Round}_1^\dagger + \text{Round}_2$)

- Generated quantum circuits are **verified**
⇒ Fix input plaintext/key and compare the ciphertext



- After verification, **quantum resources are estimated**
⇒ Qubit count, Toffoli count, and Toffoli depth



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- The LLM (GPT-5.5) receives **the guideline** together with a prompt specifying a single cipher.
 - To keep each run independent and reproducible, every circuit is generated in a temporary chat session.
- **Two types** of quantum circuits are generated: one minimizing **qubit count** (Q_{opt}) and the other **minimizing Toffoli depth** (D_{opt}).
- The implementation targets are narrowed to **lightweight block ciphers**.
(We will discuss it soon).

Results: New Quantum Implementations

- New quantum circuits generated using the LLM in this work
⇒ CRAFT, MANTIS, QARMA, mCrypton, EPCBC, and Pyjamask

Cipher	Version	Qubits	Toffoli count	Toffoli depth
CRAFT-64/128	Q opt	256	1984	124
	D opt	2720	7680	62
MANTIS5-64/128	Q opt	256	768	48
	D opt	1600	3168	24
MANTIS7-64/128	Q opt	256	1024	64
	D opt	1856	4320	32
QARMA64/128	Q opt	256	1280	80
	D opt	1888	4800	32
QARMA128/256	Q opt	512	3840	120
	D opt	4800	14720	48
mCrypton-64/64	Q opt	148	2072	182
	D opt	1456	3760	38
EPCBC-48/96	Q opt	144	4608	256
	D opt	4080	11904	64
Pyjamask-96/128	Q opt	224	1344	42
	D opt	1952	2496	14
Pyjamask-128/128	Q opt	256	1792	56
	D opt	2688	3328	28

Results: New Quantum Implementations

- Q opt $\rightarrow$ Qubit optimized version

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- **D opt** → Toffoli depth optimized version

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Results: Comparison with Previous Implementations

- Comparison with previous implementations (access to their materials is strictly limited)
⇒ LED [1], PRESENT, GIFT [2], and PRINCE [3]

Cipher	Source	Qubits	Toffoli count	Toffoli depth
LED-64	Wang et al. [22]	128	2048	128
	This paper (Q opt)	128	2048	128
	This paper (D opt)	2912	10912	64
PRESENT-80	Jang et al. [13]	144	2108	124
	This paper (Q opt)	144	2108	124
	This paper (D opt)	5628	11220	62
GIFT-64	Jang et al. [13]	192	1792	112
	This paper (Q opt)	192	1792	112
	This paper (D opt)	2688	6912	56
PRINCE	Bonnetain and Jaques [2]	192	1152	72
	This paper (Q opt)	192	960	60
	This paper (D opt)	2176	4768	24

[1] S. Wang et al. "New results in quantum analysis of led: Featuring one and two oracle attacks", Indocrypt, 2025.

[2] K. Jang et al. "Efficient implementation of PRESENT and GIFT on quantum computers", Applied Sciences, 2021.

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- **LLM-based implementation with our guideline** performs well on lightweight ciphers
⇒ The generic techniques in the guideline are sufficient
- **Limitation:** We tested our framework on AES (a more complex target), but **its performance is worse** than **state-of-the-art works**, which accumulate optimization techniques from many prior works.
⇒ The same holds for quantum circuits for RSA and ECC attacks (i.e., modular multiplication and point addition)

Current Stage of LLM-Based Quantum Implementation

Simpler targets

Lightweight block ciphers

 Moderate expert
(LLM-based implementation in this work) 

 Top expert 

Small performance gap

More complex targets

AES

RSA

ECC

 Moderate expert
(LLM-based implementation in this work) 

 Top expert 

Large performance gap

Thank you.

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